

# llmXive Automated Review of S-Agent: Spatial Tool-Use Elicits Reasoning for Spatial Intelligence

*llmXive automated review panel*

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**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

**This paper’s panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** [qwen.qwen3.5-122b](#)

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their alignment with the provided evidence.

The manuscript presents strong claims regarding the performance of S-Agent and S-Agent-8B against various baselines. While the underlying data in the tables generally supports the direction of these claims, the textual descriptions occasionally overstate the results or lack necessary precision regarding model versions and the nature of the improvements.

First, the claim in the Introduction that S-Agent-8B performs “comparably” to GPT-5.4 and Gemini 3 Pro is not strictly supported by the data. Table 3 indicates that S-Agent-8B achieves 41.6% on MMSI-Bench, which is 0.3 percentage points lower than GPT-5.4 (41.9%) and 3.6 points lower than Gemini 3 Pro (45.2%). While the performance is in the same tier, “comparable” suggests parity. The text should be revised to state that the model is “competitive with” or “approaches” these baselines to accurately reflect the observed deficits.

Second, there is a lack of precision in model naming. The Introduction mentions surpassing “GPT-5,” whereas Table 1 specifically lists “GPT-5.4.” In scientific writing, omitting the version number when multiple versions exist (e.g., GPT-5.2 vs. GPT-5.4) can lead to ambiguity about which baseline was actually beaten. All textual references to baselines should be updated to match the specific versions listed in the results tables.

Finally, the phrasing of performance gains requires clarification. The statement that S-Agent “surpasses GPT-5.4 by 4.5%” correctly reflects the absolute difference ( $46.4 - 41.9 = 4.5$ ), but in the context of benchmark results, this can be misinterpreted as a relative percentage increase. To ensure scientific rigor and prevent ambiguity, the authors should explicitly state “an absolute improvement of 4.5 percentage points.”

These issues are primarily matters of precise wording and do not invalidate the core findings, but correcting them is essential for the accuracy and credibility of the paper’s claims.

**Action items.**

- **[writing]** In the Introduction, the claim that S-Agent-8B performs ‘comparably’ to GPT-5.4 and Gemini 3 Pro is an overstatement. Table 3 shows S-Agent-8B (41.6%) is lower than GPT-5.4 (41.9%) and significantly lower than Gemini 3 Pro (45.2%). Change ‘comparably’ to ‘competitive with’ or ‘approaching’ to accurately reflect the performance gap.
- **[writing]** The Introduction claims S-Agent surpasses ‘GPT-5’ without specifying the version. Table 1 lists ‘GPT-5.4’. Ensure all model names in the text match the specific versions in the tables (e.g., GPT-5.4, not GPT-5) to avoid ambiguity and factual inaccuracy regarding the baseline used.
- **[writing]** Section 4.1 states S-Agent ‘surpasses GPT-5.4 by 4.5%’. While the absolute difference is 4.5 points, this phrasing can be ambiguous. Explicitly state ‘absolute improvement of 4.5 percentage points’ to distinguish from relative improvement and ensure scientific precision in the claim.

## Figure Critic

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_figure\_critic.md

### Figure 1

Figure 1 effectively communicates the high-level architecture of the S-Agent pipeline, but it includes two small, illegible scatter plots (in boxes m and e) that lack necessary axis labels and legends, obscuring the specific data being visualized.

### Figure 2

Figure 2 effectively illustrates the S-Agent pipeline by contrasting a failed vanilla VLM attempt with the model’s successful multi-round reasoning process. The visual layout clearly maps the progression from initial grounding failure to the final correct answer, and all text, tool outputs, and intermediate steps are legible and consistent with the caption.

### Figure 3

Figure 3 effectively illustrates the S-Agent pipeline across two distinct spatial reasoning tasks (absolute distance and object size estimation). The layout is clear, with well-defined steps, visual evidence, and final results that align with the caption’s description of qualitative visualizations.

### Figure 4

The figure effectively visualizes the S-Agent pipeline for object counting and multi-step reasoning, but the caption contradicts the document structure by claiming the figure is in the appendix, and it lacks specific descriptive details.

### Figure 5

Figure 5 presents two clear qualitative examples of the S-Agent pipeline (Relative Position and Route Planning) with well-labeled steps, tool outputs, and final answers. The visual layout is uncluttered, and the caption accurately describes the content as evidence-driven spatial reasoning.

#### Action items.

- **[writing]** Figure 1: The ‘Persistent Spatial Memory’ panel contains a small scatter plot in box (m) that is illegible and lacks axis labels or a legend, making the data it represents impossible to interpret.

- **[writing]** Figure 1: The scatter plot in box (e) ‘Spatial Reconstruction Tool’ is illegible and lacks axis labels, units, or a legend, rendering the visualization of the ‘Abs. distance’ meaningless.
- **[writing]** Figure 4: The caption states this figure is ‘in the appendix’, but the image is rendered in the main body of the preprint, creating a contradiction between the text and the document structure.
- **[writing]** Figure 4: The caption is generic (‘Additional qualitative examples’) and does not describe the specific tasks shown (Object Counting, Multi-Step Reasoning), forcing the reader to rely solely on the image headers for context.

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized terminology and acronyms that are not consistently defined for a general audience, creating barriers for non-specialist readers.

First, the acronym **VLM** (Vision-Language Model) is used repeatedly in the Introduction (Section 1) without being spelled out at the first mention. While standard in the field, a paper aiming for broad impact should define this immediately. Similarly, **SFT** (Supervised Fine-Tuning) appears in the Abstract and Introduction without definition.

Second, the term **ReAct** is introduced in the Introduction (“An LLM planner executes a ReAct loop”) without explanation. This refers to a specific reasoning framework (Reasoning + Acting) that is not self-explanatory to readers outside the agent literature. It should be briefly defined or expanded upon.

Third, in the Appendix (Section ‘Trajectory filtering’), the metric **MRA** is introduced and used in a formula without a prior definition of what the acronym stands for (Mean Relative Accuracy).

Finally, the phrase **spatio-temporal evidence accumulation** is used as a core conceptual label throughout the Abstract and Introduction. While precise, it is dense jargon. Replacing it with a more descriptive phrase like “gathering evidence across space and time” would make the core contribution more accessible without losing meaning. The authors should scan the text for other undefined acronyms (e.g., specific benchmark names like MMSI-Bench are defined, but ensure all technical terms like ‘open-vocabulary’ are contextually clear) and ensure every acronym is defined at its first occurrence.

### Action items.

- **[writing]** The manuscript relies heavily on specialized terminology and acronyms that are not consistently defined for a general audience, creating barriers for non-specialist readers. First, the acronym VLM (Vision-Language Model) is used repeatedly in the Introduction

(Section 1) without being spelled out at the first mention. While standard in the field, a paper aiming for broad impact should define this immediately. Similarly, SFT (Supervised Fine-Tuning) appears in the Abstract and Introduction without

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_logical\_consistency.md

The logical flow of the paper is generally sound, with the core premise—that spatial reasoning requires accumulating evidence over time and space rather than single-shot inference—being consistently supported by the proposed architecture and experimental results. The causal chain from the problem definition (static/stateless VLMs) to the solution (S-Agent’s memory and tool hierarchy) is clear.

However, there are minor logical inconsistencies in the presentation of quantitative results and the interpretation of ablation studies that require clarification:

1. **Ambiguity in Performance Gains:** In Section 4.2, the claim that S-Agent “improves over GPT-5.4 by 4.5%” is mathematically ambiguous. Table 1 shows an absolute difference of 4.5 percentage points (46.4 - 41.9). Using the symbol “%” without qualification often implies a relative percentage increase (which would be ~10.7%). This creates a logical disconnect between the text and the data interpretation. The text should explicitly state “4.5 percentage points” to maintain logical precision.
2. **Overgeneralization of Negative Results:** In Section 4.3, the authors state that equipping the base model with S-Agent “does not consistently improve performance” and can “hurt performance.” While Table 3 supports the “hurt” claim for MMSI-Bench (31.1% vs 30.7%), it contradicts the “hurt” implication for ViewSpatial-Bench (42.2% vs 44.1%). The logic holds that the improvement is *inconsistent*, but the phrasing “hurt performance” suggests a universal negative effect which is not supported by the full dataset. The text should be refined to specify that performance degradation was observed specifically on MMSI-Bench.
3. **Causal Attribution in Ablations:** In Section 4.4, the authors attribute the “limited benefit” of Level-2 (3D) evidence to “noisy reconstructed points” distracting the planner. While this is a plausible hypothesis, the ablation table (Table 4) only presents the aggregate accuracy drop. The logical leap from “small gain” to “noise distraction” is not directly evidenced by the numbers alone. To strengthen the logical consistency, the authors should briefly reference a qualitative failure case or a specific metric (e.g., error rate in 3D coordinate estimation) that demonstrates this noise, rather than asserting the cause without direct evidence in the results section.

These issues are primarily matters of precise phrasing and evidentiary support for causal claims

rather than fundamental flaws in the research logic.

### Action items.

- **[writing]** In Section 4.2, ‘improves over GPT-5.4 by 4.5%’ is ambiguous. Table 1 shows a 4.5 percentage point absolute gain. Clarify to ‘4.5 percentage points’ to avoid implying a relative increase.
- **[writing]** In Section 4.3, claim that tool use ‘can hurt performance’ is only supported by MMSI-Bench (31.1% vs 30.7%), not ViewSpatial (42.2% vs 44.1%). Specify the benchmark where degradation occurred to avoid overgeneralization.
- **[writing]** In Section 4.4, attribute Level-2’s limited gain to ‘noise’ without direct evidence in the ablation table. Cite a specific failure case or metric showing how noise distracts the planner to support this causal claim.

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_overreach.md

The paper makes several claims that extend beyond the immediate evidence provided in the results tables, particularly regarding “firsts” and absolute superiority over all baselines.

First, the Abstract and Introduction repeatedly claim S-Agent is the “first tool-use agent for multi-view and long-video spatial intelligence.” While the authors cite concurrent work like Think3D in the Related Work section, the main text dismisses prior art as focusing on “static images” or “single-image 3D QA.” However, Think3D (cited as [zhang2026think3d](#)) is described in the Related Work as equipping agents with “3D reconstruction and camera-manipulation tools, enabling active exploration through ego/global-view switching.” If Think3D also handles multi-view or video inputs via active exploration, the “first” claim is an overreach. The distinction between “continuous multi-view” and “active exploration” should be explicitly clarified to justify the “first” assertion, rather than broadly dismissing prior concurrent work.

Second, the claim that the distilled model S-Agent-8B “performs comparably to advanced closed-source models (e.g., GPT-5.4 and Gemini 3)” (Abstract, Introduction) is not fully supported by the quantitative results. In Table 1 (MMSI-Bench), S-Agent-8B scores 41.6%, while GPT-5.4 scores 41.9%. In Table 3 (ReVSI), S-Agent-8B scores 52.8%, while Gemini 3 Pro scores 60.9%. While the performance is competitive, stating it is “comparable” to models that consistently outperform it by 1-8 percentage points on key benchmarks is an overstatement. The language should be adjusted to “competitive with” or “approaching the performance of” to accurately reflect the data.

Third, the Abstract states that S-Agent “surpasses... all open-weight spatial models.” Table 1 shows S-Agent (46.4%) outperforming the listed open-weight spatial models (e.g., SN-SI-1.1-Qwen3VL-8B at 38.1%, VST-7B-SFT at 32.5%). However, the Related Work section and Table 1

caption mention SN-SI-1.3-IVL3-8B, which achieves 61.3% on ViewSpatial-Bench (a benchmark where S-Agent scores 60.0%). If SN-SI-1.3-IVL3-8B is considered an “open-weight spatial model” (which the context implies), the claim of surpassing “all” such models is factually incorrect on the ViewSpatial benchmark. The claim should be qualified to specify which benchmarks or which subset of models are surpassed.

Finally, the claim in the Introduction that “Simply and directly applying the S-Agent framework consistently improves the spatial reasoning ability of these VLMs” is slightly overreaching. Table 2 shows that for the Qwen3-VL-8B backbone, the zero-shot S-Agent variant (30.7% on MMSI) actually performs *worse* than the base Qwen3-VL-8B-Instruct (31.1%). The text acknowledges this in Section 4.2 (“simply equipping the base Qwen3-VL-8B-Instruct with S-Agent does not consistently improve performance”), but the Introduction’s phrasing suggests a universal improvement that the data contradicts for specific base models. The Introduction should be revised to reflect that improvements are consistent for *stronger* planners (like GPT-5.4) or that the framework *enables* improvement which is then realized through distillation, rather than implying the zero-shot framework alone always improves performance.

#### Action items.

- **[writing]** The claim that S-Agent is the ‘first tool-use agent for multi-view and long-video spatial intelligence’ (Abstract, Intro) is overreaching given the existence of concurrent works like Think3D (cited in Related Work) which also handles active 3D exploration. The distinction needs to be more precise regarding ‘continuous’ vs ‘active exploration’ to avoid hyperbole.
- **[writing]** The statement that S-Agent-8B ‘performs comparably to advanced closed-source models (e.g., GPT-5.4 and Gemini 3)’ (Abstract, Intro) is not fully supported by the data. Table 1 shows S-Agent-8B (41.6%) trailing GPT-5.4 (41.9%) on MMSI-Bench and trailing Gemini 3 Pro (60.9%) on ReVSI. The claim should be tempered to ‘competitive with’ or ‘approaching’ rather than ‘comparable to’ given these specific deficits.
- **[writing]** The claim that S-Agent ‘surpasses... all open-weight spatial models’ (Abstract) is contradicted by Table 1 and Related Work, where SN-SI-1.3-IVL3-8B scores 61.3% on ViewSpatial-Bench, outperforming S-Agent’s 60.0%. The claim of surpassing ‘all’ is inaccurate on this benchmark and needs qualification.

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_safety\_ethics.md

The manuscript presents S-Agent, a framework for spatial reasoning using tool-augmented agents. From a safety and ethics perspective, the work is generally sound as it focuses on benchmark evaluation rather than direct deployment in high-risk physical environments. However, several

areas require clarification to ensure responsible dissemination and usage of the resulting models and datasets.

First, the construction of the **S-300K** dataset (Section 4.1, Appendix B) involves using a proprietary model (GPT-5.4) to generate trajectories from the **SenseNova-SI-800K** dataset. The paper states that SenseNova-SI-800K is “fully disjoint” from evaluation benchmarks but does not explicitly detail the licensing terms of the source data or the resulting distilled dataset. Given the potential for copyright or licensing conflicts when distilling proprietary model outputs into open-weight datasets, the authors must explicitly state the license under which S-300K is released and confirm that the use of SenseNova-SI-800K permits such derivative works. This is critical for the reproducibility and legal safety of downstream users.

Second, the system relies heavily on external tools like **GroundingDINO** (Appendix A) for object detection and **Depth-Anything-3** for metric depth. These foundational models are known to exhibit biases (e.g., under-detection of certain object classes or demographic groups) and failure modes in specific lighting or environmental conditions. While the paper focuses on spatial reasoning accuracy, it should briefly acknowledge how these underlying tool limitations could introduce safety risks if the agent were deployed in real-world scenarios (e.g., autonomous driving or robotics). A short discussion on the propagation of these biases or the potential for “hallucinated” geometric evidence due to tool errors would strengthen the ethical robustness of the work.

Finally, the Introduction (Section 1) highlights applications in **autonomous driving** and **embodied robotics**. While the current evaluation is limited to static benchmarks (MMSI-Bench, ReVSI, etc.), the authors should include a clear disclaimer or limitation statement clarifying that the current system has not been validated for safety-critical physical deployment. This prevents potential misinterpretation of the benchmark results as a guarantee of safety in real-world, dynamic environments.

No fatal ethical violations or dual-use risks (e.g., generating harmful content, surveillance capabilities) were identified in the current scope. The primary concerns are related to data licensing transparency and the responsible communication of system limitations.

#### **Action items.**

- **[writing]** The paper describes generating 300K training trajectories using GPT-5.4 (a proprietary API) on the SenseNova-SI-800K dataset. Explicitly state the data licensing terms of SenseNova-SI-800K and confirm that the resulting S-300K dataset is released under a compatible license (e.g., CC-BY, MIT) to ensure downstream users can legally utilize the distilled agent.
- **[writing]** The methodology relies on open-vocabulary object detection (GroundingDINO) and metric depth estimation. Add a brief discussion on potential biases in these tools (e.g., demographic bias in detection, failure modes in specific environments) and how they might propagate into the agent’s spatial reasoning, particularly for safety-critical applications like robotics.
- **[writing]** The paper mentions applications in autonomous driving and embodied robotics. Include a specific statement on the limitations of the current evaluation (benchmarks)

regarding real-world safety risks, and clarify that the system is not yet validated for deployment in safety-critical physical environments.

## Scientific Evidence

*Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_scientific\_evidence.md

The scientific evidence presented for S-Agent is generally strong, demonstrating consistent improvements over baselines across multiple benchmarks (MMSI-Bench, ViewSpatial-Bench, ReVSI). The ablation studies in Section 4.4 (Table 2) effectively isolate the contributions of the hierarchical evidence levels and memory modules, showing that Level-3 experts and memory are critical for performance. The qualitative analysis in Section 4.5 provides concrete examples of how the agent corrects errors through tool use, supporting the central claim of evidence accumulation.

However, there are three specific concerns regarding the robustness and interpretation of the evidence:

1. **Statistical Robustness of Ablations:** The ablation study (Table 2, `tab:ablation_config`) reports single-point accuracy scores for each configuration (e.g., 56.7% for “Spatial only” vs. 58.2% for “+ Scene memory”). Given the small magnitude of some gains (e.g., 1.5% from scene memory), it is essential to know if these improvements are statistically significant. The paper does not report standard deviations from multiple random seeds or any significance testing (e.g., t-tests). Without this, it is difficult to rule out that the observed gains are due to random variance in the evaluation or the specific initialization of the planner.
2. **Overstated Comparability to Closed-Source Models:** The abstract and introduction claim that S-Agent-8B “performs comparably” to advanced closed-source models like GPT-5.4 and Gemini 3 Pro. While S-Agent-8B outperforms the base Qwen3-VL-8B significantly, the data in Table 1 (`tab:trajectory_distillation`) shows a notable gap on the ReVSI benchmark (52.8% for S-Agent-8B vs. 60.9% for Gemini 3 Pro). On MMSI-Bench, the gap is smaller (41.6% vs. 45.2%), but the claim of “comparability” across the board is an overstatement given the 8-point deficit on ReVSI. The authors should refine this claim to reflect the specific areas of parity and the areas where a gap remains.
3. **Teacher-Student Bias in Distillation:** The S-300K dataset is constructed by generating trajectories with a GPT-5.4 teacher and filtering for correctness. This methodology risks the student model (S-Agent-8B) overfitting to the specific reasoning heuristics, tool selection strategies, and potential biases of the GPT-5.4 teacher. If the teacher fails on a specific type of spatial query or uses a suboptimal tool sequence that happens to yield the correct answer, the student may learn this suboptimal behavior. The paper does not discuss the potential for this “teacher bias” to limit the generalization of the distilled agent to scenarios where the

teacher’s strategy is not optimal or where a different planner might be used. A discussion on the robustness of the distilled policy to variations in the teacher or the inclusion of diverse reasoning paths would strengthen the evidence.

### Action items.

- **[science]** The ablation study in Table 2 (tab:ablation\_config) lacks statistical significance testing. With only a single run reported for each configuration (e.g., 56.7% vs 58.2%), it is unclear if the gains from memory modules are robust or due to random variance. Please report standard deviations over multiple seeds or perform a significance test.
- **[science]** The claim that S-Agent-8B ‘performs comparably’ to GPT-5.4 and Gemini 3 Pro (Abstract, Intro) is not fully supported by the data in Table 1 (tab:trajectory\_distillation). On ReVSI, S-Agent-8B scores 52.8% while Gemini 3 Pro scores 60.9%, a gap of 8.1 points. The authors should qualify this claim or provide a more nuanced comparison rather than asserting comparability across all metrics.
- **[science]** The training data construction (S-300K) relies on a ‘frozen teacher’ (GPT-5.4) to generate trajectories, which are then filtered for correctness. This introduces a potential bias where the student model learns to mimic the specific reasoning patterns and tool preferences of the GPT-5.4 teacher, potentially limiting generalization to other planners or scenarios where the teacher fails. The authors should discuss this teacher-student bias and its impact on the robustness of the distilled agent.

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_statistical\_analysis.md

The manuscript presents a compelling framework for spatial reasoning, but the statistical rigor of the experimental evaluation requires strengthening to support the strong claims of superiority over baselines.

First, the ablation studies in Section 4.3 and Table 2 (tab\_ablation\_config) report performance improvements for individual modules (e.g., adding Scene Memory yields +1.5% over the spatial-only baseline) as single point estimates. Given the stochastic nature of LLM planning and potential variance in tool outputs (e.g., depth estimation noise), these marginal gains could be within the noise floor of the evaluation. The authors should report results over multiple random seeds (e.g., 3-5) to calculate standard deviations or 95% confidence intervals. Without this, it is impossible to determine if the observed improvements are statistically significant or merely artifacts of a specific random seed or evaluation split.

Second, the main results in Tables 1, 2, and 3 claim that S-Agent “surpasses” or is the “best” based on small accuracy differences (e.g., 46.4% vs 45.2% on MMSI-Bench, a 1.2% gap). In benchmark evaluations, especially with complex agent loops, variance can be substantial. The

paper lacks any form of statistical significance testing (e.g., paired t-tests, Wilcoxon signed-rank tests, or bootstrap confidence intervals) to validate these claims. A difference of 1.2% is often not statistically significant in NLP/CV benchmarks without rigorous error analysis. The authors must provide variance metrics or significance tests to substantiate that their method is genuinely superior rather than just slightly better on a specific run.

Finally, regarding the data construction in Appendix A.2, the filtering of numeric trajectories relies on a hard threshold of Mean Relative Accuracy (MRA)  $\geq 0.6$ . The choice of this specific threshold appears arbitrary without statistical justification. A sensitivity analysis showing how the final model performance varies with different MRA thresholds (e.g., 0.5, 0.6, 0.7) would strengthen the reproducibility and robustness of the data generation pipeline. Currently, the lack of variance reporting and significance testing limits the confidence in the reported state-of-the-art claims.

### Action items.

- **[science]** In Section 4.3 (Ablation Studies) and Table 2 (tab\_ablation\_config), the performance gains of individual components (e.g., +2.5% for Scene Memory) are reported as single point estimates. Given the iterative nature of the agent and potential variance in tool execution, please report confidence intervals or standard deviations (e.g., via 3-5 random seeds) to establish statistical significance of these marginal gains.
- **[science]** In Section 4.2 (Zero-Shot Performance) and Tables 1-3, the paper claims ‘best overall’ or ‘surpasses’ baselines based on point-estimate accuracy differences (e.g., 46.4% vs 45.2%). Without reported variance or significance testing (e.g., paired t-tests or bootstrap confidence intervals), it is unclear if these small margins (1.2%) are statistically significant or within the noise of the evaluation protocol.
- **[science]** In Appendix A.2 (Details of S-300K), the trajectory filtering criteria for numeric questions uses a fixed threshold of Mean Relative Accuracy (MRA)  $\geq 0.6$ . The paper does not justify this threshold statistically or discuss its sensitivity. Please provide a sensitivity analysis or justification for this cutoff to ensure it does not introduce selection bias in the training data.

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_writing\_quality.md

The manuscript demonstrates a high standard of academic writing, with clear articulation of the proposed S-Agent framework and its motivation. The logical flow from the problem statement (limitations of single-shot VLMs) to the solution (agentic evidence accumulation) is well-structured and easy to follow. The use of formal notation in the Method section is consistent and aids in defining the agent’s state transitions clearly.

However, there are minor areas where sentence-level conciseness and transitional flow can be improved. In the Introduction, some sentences are slightly verbose due to the stacking of adverbs (e.g., “Simply and directly applying...”). While the meaning remains clear, tightening these phrases would enhance the professional tone. Additionally, in the Experiments section, the reporting of results occasionally repeats the comparison logic in close proximity (e.g., stating the score and then immediately the percentage improvement in a way that feels slightly redundant).

The transition between the formal mathematical formulation of the memory update mechanism and the subsequent descriptive paragraphs in the Method section could be smoother. Currently, the text jumps from equations to a bolded term without a bridging sentence that explicitly guides the reader on how the formal definition maps to the specific memory components (Scene vs. Agent Memory).

Overall, the writing is strong and effectively communicates the technical contributions. The suggested revisions are minor and focus on polishing the prose for maximum clarity and impact.

#### **Action items.**

- **[writing]** In Section 1 (Introduction), the sentence ‘Simply and directly applying the S-Agent framework consistently improves...’ contains redundant adverbs. ‘Simply’ and ‘directly’ convey similar meanings here; consider removing one to improve conciseness and flow.
- **[writing]** In Section 4 (Experiments), the phrase ‘improving over GPT-5.4 by 4.5% on MMSI-Bench’ appears immediately after stating the absolute score. The phrasing is slightly repetitive. Consider restructuring to: ‘...achieving 46.4%, a 4.5% absolute improvement over GPT-5.4.’
- **[writing]** In Section 3 (Method), the transition between the formal definition of memory updates and the description of Scene Memory is abrupt. The paragraph starting ‘Scene Memory turns 2D/3D cues...’ begins with a bolded term but lacks a clear introductory sentence linking it to the previous formal equations.
- **[writing]** In the Appendix (Section 7), the description of the ‘Metric measurement expert’ uses the phrase ‘deterministically maps the request to a measurement route’. The word ‘deterministically’ is used frequently throughout the paper; ensure it is necessary here or if ‘systematically’ or ‘explicitly’ would be more precise given the context of tool selection.